One grid. Every tile configurable.

MOSAIC flat architecture · one R × C mesh · example placement includes all five tile types

Host processor + software
Choose hardware, program contexts and routes
Launch work and collect results through the adapter

Commands · responses · events

System memory + visibility service
Application inputs, outputs and configuration images
Platform-defined memory access and synchronization

Platform memory / data service

Clock, reset + test sources
One common functional clock / reset domain
Optional shared gate controls fabric clock activity

Clock / reset / test

MOSAIC · single-grid CGRA

R × C tile positions | Independent FUs per tile | Example: 2 × 3 tiles

Host-facing integration

Dashed group: proposed platform adapter

01 Configuration + routes
Write / validate tile context images
Program each local switch through its native ports

02 Runtime + context control
Allocate / start / suspend / drain native tiles
Activate committed contexts at safe points

03 Memory + stream service
Host SRAM access and native MST DMA streams
A platform adapter connects external memory

04 Events + observability
Collect completion / error records and status
Expose counters and platform interrupt policy

Grid exports native per-tile and route interfaces.
Host transport is selected by platform integration.

config

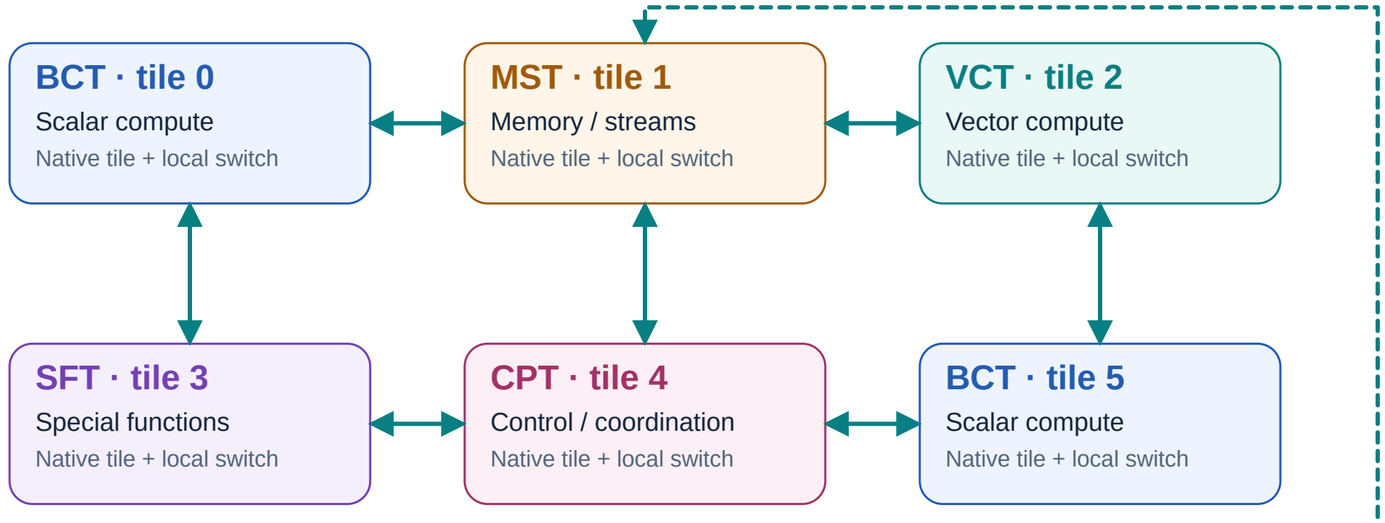
runtime

data

events

05 Single R × C tile mesh

Neighbor links carry data through statically configured local switches.



Perimeter streams connect external data producers and consumers.

MST memory bridge

06 Shared banked scratchpad
8 banks × 32 words × 8 B = 2,048 B · host client + one memory client per MST

Common mesh payload width follows the widest selected native tile.

Blue: runtime/control | Purple: configuration | Teal: data/memory | Amber: events/status | Arrows summarize native interface bundles

FROM A HARDWARE FLAVOR TO EXECUTING DATAFLOW

1 Choose hardware
Select dimensions, tile placement, FUs and private widths / storage settings.

→

2 Program the grid
Write / validate contexts, configure routes and memory permissions.

→

3 Execute
Launch the native tiles; stream data through the routed neighbor mesh.

→

4 Collect results
Store outputs and observe completion. Software synchronizes external buffers.

All tile positions belong to the same mesh. Hardware selection and runtime context programming are separate steps.

Grid and native tiles follow the Configurable backend. The dashed host-facing adapter is an integration concept.
Placement is logical connectivity. Silicon area, timing, power and energy require later technology-specific implementation.

Build the CGRA around your workload

Local Configurable backend · independent rows and columns · any tile class at any position

Grid + placement

Choose rows × columns and every tile class / ID.
Coordinates determine neighbors and hop distance.

Private hardware per tile

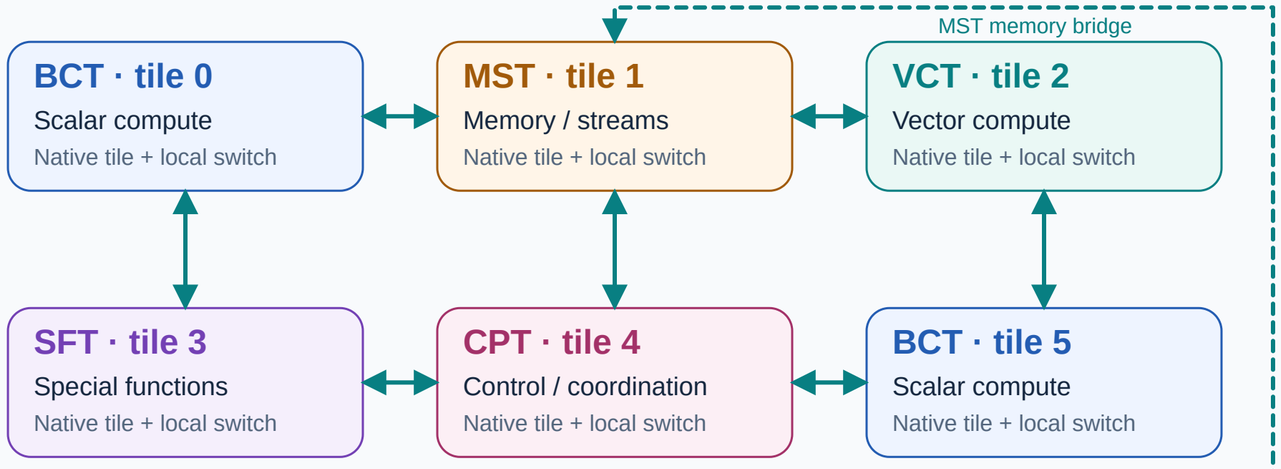
Choose FUs, widths, lanes, storage and protection.
Settings belong to that individual tile instance.

Shared memory + clocks

Choose SRAM banking / depth and clock gating.
One synchronous domain; four contexts per tile.

Example fabric · one 2 × 3 tile grid

2 × 3 example from app/examples/heterogeneous.toml; dimensions are not presets.



Edge streams and each tile's native control ports remain exposed.

Shared banked scratchpad

8 banks × 32 words × 8 B = 2,048 B · one host client + one client per MST

Inside every native tile

Native configuration / runtime / predicate / event ports

07 Tile shell · shared contract

Context memory + safe-point context selection
Owned queues + lifecycle and identity checks
Monitoring + power-control handshakes
Protection according to the selected native parameters

08 Tile-specific body

Selected FUs + registers, accumulators and local state
Unavailable operation classes are rejected

Data streams connect through the separate local switch.

A shell power protocol does not imply enabled physical power gating.

09 / CHOOSE EACH FUNCTIONAL UNIT INDEPENDENTLY

BCT
Basic compute tile

Scalar ALU
Multiplier
MAC

Scalar integer dataflow

VCT
Vector compute tile

Lane ALU
Lane multiply · lane MAC
Vector reduction
Dot product

Masked SIMD + scalar results

SFT
Specialized function tile

ALU · multiplier
LUT · polynomial
Reciprocal · RSQRT · exponential
Reduction · formatter
Accumulator

Fixed-point nonlinear functions

MST
Memory + stream tile

Load engine
Store engine
AGUs + formatters live
inside the selected engines

SRAM / stream movement

CPT
Control + predicate tile

Predicate operations
Loop controller
Events / barriers
Tagged actions

Control + synchronization

WHAT A CONFIGURATION CHANGES

Widths + lane count

Change number range, payload size and replicated logic.
The widest tile sets the common grid payload width.

Tables, queues + latency

Change capacity, storage and backpressure behavior.
A response delay does not necessarily retime arithmetic.

Hardware versus runtime contexts

Generation determines which hardware exists.
Contexts program operations within that hardware.

Native per-tile controls connect to a host adapter; mesh edge streams connect external producers and consumers.
Layout is logical connectivity, not a floorplan. Area, timing, power and energy require later technology-specific implementation.